

AICE3001/AICE6002: Mathematics of Machine Learning

Subsidiary Notes — Lecture 5: Inner Product Spaces

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1 Keywords

- Inner-product, operators

2 Main Points

- For some vector spaces we can define an *inner product* between pairs of vectors
- Inner products are scalars associated with two elements in a vector space
- They are generally denoted by $\langle \mathbf{x}, \mathbf{y} \rangle$
- To be an inner-product requires satisfying 5 axioms

1. $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$ for all $\mathbf{x} \in \mathcal{V}$
2. $\langle \mathbf{x}, \mathbf{x} \rangle = 0$ if and only if $\mathbf{x} = \mathbf{0}$
3. $\langle \alpha \mathbf{x}, \mathbf{y} \rangle = \alpha \langle \mathbf{x}, \mathbf{y} \rangle$
4. $\langle \mathbf{x}, \mathbf{y} + \mathbf{z} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{x}, \mathbf{z} \rangle$
5. $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$

- The inner-product induces a norm: $\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$
- For normal vectors (i.e. $\mathbf{x} \in \mathbb{R}^n$) the standard inner product is the dot-product

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y} = \sum_{i=1}^n x_i y_i$$

- We can define an inner product between functions

$$\langle f, g \rangle = \int_{x \in \mathcal{I}} f(x) g(x) dx$$

- For matrices we can define the Frobenius inner-product

$$\langle \mathbf{A}, \mathbf{B} \rangle_F = \text{Tr } \mathbf{A}^\top \mathbf{B} = \sum_{i,j} A_{ij} B_{ij}$$

which defines the Frobenius norm

$$\|\mathbf{A}\|_F = \sqrt{\langle \mathbf{A}, \mathbf{A} \rangle_F} = \sqrt{\sum_{i,j} A_{ij}^2}$$

The Frobenius norm is not a compatible norm.

- A really important inequality is the *Cauchy-Schwarz* inequality

$$\langle \mathbf{x}, \mathbf{y} \rangle^2 \leq \langle \mathbf{x}, \mathbf{x} \rangle \langle \mathbf{y}, \mathbf{y} \rangle = \|\mathbf{x}\|^2 \|\mathbf{y}\|^2$$

- Inner products allow us to define the notion of similarity

$$\begin{aligned} \langle \mathbf{x}, \mathbf{y} \rangle &= \|\mathbf{x}\| \|\mathbf{y}\| \cos(\theta) \\ \langle f(x), g(x) \rangle &= \|f(x)\| \|g(x)\| \cos(\theta) \end{aligned}$$

- $\cos(\theta)$ can be seen as a measure of the correlation between vectors (or functions)
- Because of Cauchy-Schwarz $\langle \mathbf{x}, \mathbf{y} \rangle / (\|\mathbf{x}\| \|\mathbf{y}\|)$ lies between -1 and 1 (so that we can represent this quantity by the cosine of an angle)

2.1 Coordinates or Basis Vectors

- Any set of vectors that span the entire vector space can be considered a set of basis vectors or coordinates
- If our bases are linearly independent then we have a set of non-degenerate basis function where each vector is unique
- The most convenient set of basis vectors are those where the bases are normalised and orthogonal $\langle \mathbf{b}_i, \mathbf{b}_j \rangle = \delta_{ij}$

- **Basis Functions**

- For a function space we can consider a set of basis functions
- A familiar set of functions define on the interval $[0, 2\pi]$ are the Fourier functions

$$\{1, \cos(\theta), \sin(\theta), \cos(2\theta), \sin(2\theta), \dots\}$$

- This basis set is orthogonal as for any two components $\langle b_i(\theta), b_j(\theta) \rangle = \delta_{ij}$
- There are many orthogonal polynomials that have similar properties
- Given an orthogonal set of functions $\{b_i(\mathbf{x})\}$ we can decompose a function $f(\mathbf{x})$ as a (infinite) vector $\mathbf{f}$ with components $f_i = \langle f(\mathbf{x}), b_i(\mathbf{x}) \rangle$
- This allows us to represent any function as a point in an infinite-dimensional space

2.2 Operators

- Operators transform elements of a vector space
- Consider the transformation or operator $\mathcal{T} : \mathcal{V} \rightarrow \mathcal{V}'$
- This says that $\mathcal{T}$ maps some object $\mathbf{x} \in \mathcal{V}$ to an object $\mathbf{y} = \mathcal{T}[\mathbf{x}]$ in a new vector space $\mathcal{V}'$
- **Linear Operators**

- Linear operators satisfy the two conditions
 1. $\mathcal{T}[a \mathbf{x}] = a \mathcal{T}[\mathbf{x}]$
 2. $\mathcal{T}[\mathbf{x} + \mathbf{y}] = \mathcal{T}[\mathbf{x}] + \mathcal{T}[\mathbf{y}]$
- Linear operators are really important because we can understand them
- For normal vectors the most general linear operation is

$$\mathcal{T}[\mathbf{x}] = \mathbf{M} \mathbf{x}$$

where $\mathbf{M}$ is a matrix

- For functions the most general linear operator is a kernel function

$$g(\mathbf{x}) = \mathcal{T}[f(\mathbf{x})] = \int K(\mathbf{x}, \mathbf{y}) f(\mathbf{y}) d\mathbf{y}$$

- Kernels appear in SVMs, SVRs, kernel-PCA, Gaussian Processes
- Often we are interested in operators that map vectors in a vector space to new vectors in the same vector space
 - $\mathcal{T} : \mathcal{V} \rightarrow \mathcal{V}$
 - The most general linear mapping for normal vectors that does this are square matrices

2.3 Matrices and Mappings

- Matrices, $\mathbf{M}$ are linear maps from one point $\mathbf{x}$ to another $\mathbf{y} = \mathbf{M}\mathbf{x}$
- The product of a matrix $\mathbf{C} = \mathbf{A}\mathbf{B}$ corresponds to applying the mapping $\mathbf{B}$ followed by $\mathbf{A}$
- For most matrices $\mathbf{A}\mathbf{B} \neq \mathbf{B}\mathbf{A}$ (we say matrix multiplication does not commute)
- There are pairs of matrices that do commute, but they need to share a special structure for this to happen
- Matrix multiplication is associative. That is, $(\mathbf{A}\mathbf{B})\mathbf{C} = \mathbf{A}(\mathbf{B}\mathbf{C})$